

CS-233 Introduction to Machine Learning — Milestone 1 Report

1 Introduction

In this milestone, we implemented and evaluated several machine learning methods on the provided Gaming and Mental Health dataset. We considered two prediction tasks: a classification task, where the goal is to predict the addiction level, and a regression task, where the goal is to predict the addiction score. We implemented K-Nearest Neighbors (KNN) for both tasks, Logistic Regression for classification, and Linear Regression for regression. Our objective was not only to compare predictive performance, but also to analyze generalization, overfitting, and computational efficiency across methods.

2 Method

2.1 Preprocessing and validation protocol

All input features were standardized using

$$x' = \frac{x - \mu}{\sigma},$$

where μ and σ were estimated only on the training split. The same transformation was then reused on the validation and test data. If a feature had zero standard deviation, we set its standard deviation to 1 to avoid numerical issues. For model selection, we used an 80/20 train-validation split. The validation set was used to choose the hyperparameters, and the final selected models were then evaluated on the test set.

2.2 Models

KNN. We used KNN for both classification and regression, using Euclidean distance to find the nearest neighbors. For classification, the predicted class is the majority label among the K nearest neighbors. For regression, the prediction is the average target value of these neighbors.

Logistic Regression. For classification, we implemented a multi-class Logistic Regression model with a softmax output, optimized with gradient descent.

Linear Regression. For regression, we implemented Linear Regression using the closed-form solution with an explicit bias term.

During validation, we explored $K \in \{1, 3, 5, 7, 11, 15\}$ for KNN and learning rates in $\{0.001, 0.01, 0.05\}$ with `max_iters` in $\{500, 1000\}$ for Logistic Regression. The final retained settings were $K = 3$ for KNN classification, $K = 11$ for KNN regression, and learning rate 0.05 with 1000 iterations for Logistic Regression.

3 Results

3.1 Final test performance

Classification

Method	Accuracy	Macro-F1
KNN ($K = 3$)	76.50%	0.552
Logistic Regression	85.75%	0.597

Regression

Method	MSE
KNN ($K = 11$)	1.880
Linear Regression	0.994

On the final test set, Logistic Regression was the strongest model for classification, outperforming KNN on both reported

metrics. For regression, Linear Regression achieved a substantially lower MSE than KNN, making it the best-performing method for predicting the addiction score.

3.2 Hyperparameter analysis

The validation experiments showed that the choice of K was important for KNN. In classification, small values of K gave very strong training performance but weaker validation behavior, which is a sign of overfitting. In our experiments, $K = 3$ gave the best trade-off between accuracy and macro-F1. In regression, increasing K reduced the influence of individual noisy neighbors and improved the validation MSE up to around $K = 11$, after which the gain became marginal. For Logistic Regression, the best validation setting was obtained with learning rate 0.05 and 1000 iterations.

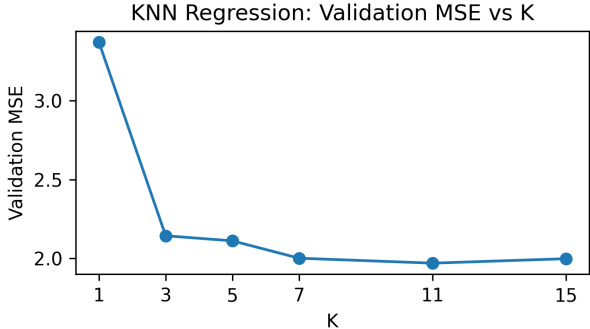


Figure 1: Validation MSE of KNN regression as a function of K . The largest improvement occurs when moving away from $K = 1$, while performance stabilizes near $K = 11$, which was retained for the final test run.

3.3 Runtime analysis

Method	Fit time (s)	Predict time (s)
KNN Classification	0.121	0.030
KNN Regression	0.115	0.029
Logistic Regression	0.068	0.00005
Linear Regression	0.0004	0.00001

From a computational point of view, Linear Regression was the cheapest model both to train and to evaluate. Logistic Regression also remained efficient, whereas KNN was clearly the most expensive at prediction time because each test point requires comparisons with the entire training set.

4 Discussion

The results show that the best method depends on the prediction task. For classification, Logistic Regression outperformed KNN on both accuracy and macro-F1, suggesting that a learned parametric decision rule was more effective here than relying only on local neighbors. However, the macro-F1 remained lower than the accuracy, which suggests that performance was not equally strong across all classes.

For regression, Linear Regression clearly outperformed KNN and also had a lower prediction cost. This suggests that the addiction score can already be approximated reasonably well with a simple linear model on these features. KNN remained useful as a comparison method, but it was more sensitive to the choice of K and slower at prediction time.

The validation experiments also highlighted the importance of hyperparameter tuning. In particular, small values of K in KNN led to signs of overfitting, while larger values produced more stable predictions. For Logistic Regression, the learning rate and number of iterations also mattered, since weaker optimization settings gave worse validation results.

5 Conclusion

To conclude, Logistic Regression obtained the best classification results, while Linear Regression obtained the lowest MSE for regression. KNN was simple and useful for comparison, but it was more sensitive to K and slower at prediction time. Overall, the experiments showed that validation results, test performance, and runtime all matter when comparing models.